

Institution's Innovation Council (IIC) Portal

Complete Project Brief & Development Documentation

1. Project Overview

The **Institution's Innovation Council (IIC) Portal** is a modern, responsive web application designed to serve as the official digital platform of the Institution's Innovation Council.

The portal will showcase the institution's innovation activities, leadership, events, notices, research, active projects, certifications, and Innovation Ambassadors. It will also provide an interactive platform where students and users can **submit innovative ideas and ask queries**.

An **Admin Dashboard** will allow authorized administrators to manage the website content without modifying the source code.

The application will use **React** for the frontend and **Supabase** for database, authentication, and file storage. There will be **no traditional Node.js/Express backend**.

2. Project Objectives

The main objectives are:

- Create a modern official website for the IIC.
- Showcase IIC activities and achievements.
- Display leadership and team members.
- Publish upcoming and past events.
- Provide live countdowns for upcoming events.
- Publish official notices and announcements.
- Showcase Innovation Ambassadors.
- Display research and development activities.
- Showcase active student and institutional projects.
- Display establishment documents and certifications.
- Allow users to submit innovative ideas.
- Allow users to submit queries.
- Provide a secure Admin Dashboard.

- Allow administrators to manage website content dynamically.
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3. Technology Stack

Frontend

- **React**
- **Vite**
- **React Router DOM**
- **Tailwind CSS**
- **Framer Motion**
- **Lucide React**
- **React Hook Form**
- **React Toastify**
- **Swiper.js**

Backend Services

Supabase

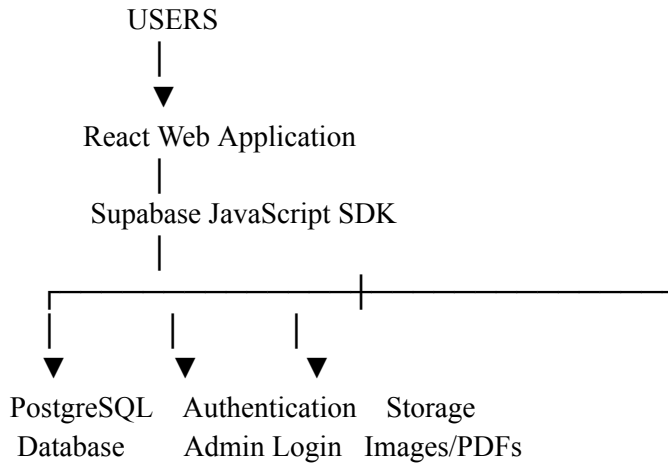
Supabase will provide:

- PostgreSQL Database
- Admin Authentication
- File Storage
- Row Level Security (RLS)

Deployment

- **Frontend:** Vercel or Netlify
 - **Database:** Supabase PostgreSQL
 - **Authentication:** Supabase Auth
 - **Images and Documents:** Supabase Storage
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4. System Architecture



There is no separate Express or Node.js backend.

5. User Roles

5.1 Public User

Public users can:

- View all public pages.
- View leadership and members.
- View events.
- View notices.
- View Innovation Ambassadors.
- View research activities.
- View active projects.
- View certifications.
- Submit ideas.
- Submit queries.
- Submit contact messages.

Public users cannot:

- Access the Admin Dashboard.
 - Edit or delete website content.
-

5.2 Administrator

Administrators can:

- Securely log in.
 - Access the Admin Dashboard.
 - Add, edit, and delete website content.
 - Manage leadership.
 - Manage members.
 - Manage events.
 - Manage notices.
 - Manage Innovation Ambassadors.
 - Manage R&D content.
 - Manage projects.
 - Manage certifications.
 - Review submitted ideas.
 - View and reply to queries.
 - View contact messages.
 - Manage website settings.
-

6. Website Pages

The website will contain the following main pages:

Home

- About
 - Leadership & Members
 - Events
 - Notices
 - Innovation Ambassadors
 - Research & Development
 - Active Projects
 - Establishment & Certifications
 - Ideas & Queries
 - Contact Us
 - Admin Dashboard
-

7. Home Page

The Home page will be the main landing page of the website.

Sections

Hero Section

Contains:

- IIC name
- Institution name
- Main tagline
- Short introduction
- Call-to-action buttons
- Modern animated visual

Example:

Innovate. Inspire. Impact.

About Preview

A short introduction to the IIC with a **Learn More** button.

Statistics

Display animated statistics such as:

- Events Conducted
- Active Projects
- Innovation Ambassadors
- Ideas Received
- Research Projects

Upcoming Event

Display the next upcoming event with:

- Event poster
- Event title
- Date
- Venue

- Live countdown
- Registration button

Latest Notices

Display the latest 3–5 notices.

Each notice shows:

- Title
- Date
- Category
- NEW badge
- View details

Featured Projects

Display selected active projects.

Innovation Highlights

Show important achievements and activities.

Gallery Preview

Display recent activity images.

Contact CTA

Encourage users to:

- Submit an idea
- Ask a query
- Contact the IIC

8. About Page

The About page provides information about the Institution’s Innovation Council.

Sections

- About IIC
 - History
 - Vision
 - Mission
 - Objectives
 - Functions of IIC
 - Institution Details
-

9. Leadership & Members Page

This page displays the people responsible for managing the IIC.

Leadership Section

Important leadership profiles include:

- Director
- Chairperson
- Faculty Coordinator
- President
- Startup Coordinator
- Other key leadership positions

Each profile contains:

- Photo
- Name
- Position
- Department
- Short description
- Email, if required
- LinkedIn, if required

Members Section

Members can be grouped according to teams.

Examples:

- Executive Team

- Technical Team
- Media Team
- Design Team
- Event Team
- Documentation Team
- Other Teams

Each member contains:

- Photo
 - Name
 - Position
 - Team
 - Department
-

10. Events Page

The Events page will have two sections.

Upcoming Events

Each event contains:

- Event poster
- Event title
- Description
- Date
- Time
- Venue
- Registration link
- Live countdown

Example:

INNOVATION WORKSHOP

Starts In:

04 Days : 12 Hours : 35 Minutes

Past Events

Past events display:

- Event title
- Poster
- Date
- Description
- Gallery
- Event report
- Winners, if applicable

Events automatically move from upcoming to past based on the event date, where appropriate.

11. Notices Page

The Notices page displays official IIC announcements.

Types of Notices

- General
- Important
- Events
- Workshops
- Competitions
- Hackathons
- Funding Opportunities
- Internship Opportunities

Each Notice Contains

- Notice title
- Description
- Category
- Published date
- Deadline
- PDF attachment, if available
- External link, if available
- NEW badge for recent notices

Additional Features

- Search notices
- Filter by category
- Sort by latest
- Pin important notices

The Home page will display the latest notices.

12. Innovation Ambassadors Page

This page showcases the institution's Innovation Ambassadors.

Each profile contains:

- Photo
 - Name
 - Department
 - Position
 - Responsibilities
 - Achievements
 - Year/Batch
-

13. Research & Development Page

The R&D page showcases research and innovation activities.

Sections

- Ongoing Research
- Completed Research
- Publications
- Patents
- Research Collaborations
- Research Achievements

Each research item can contain:

- Title
 - Description
 - Researcher/Team
 - Faculty Mentor
 - Category
 - Status
 - Images
 - Documents
-

14. Active Projects Page

This page showcases projects currently being developed.

Each project contains:

- Project title
- Cover image
- Description
- Team members
- Team lead
- Faculty mentor
- Technologies used
- Category
- Current status
- Progress
- GitHub or project link, if public

Project Categories

Examples:

- Artificial Intelligence
- Machine Learning
- IoT
- Robotics
- Web Development
- Cybersecurity
- Sustainability
- Healthcare
- Other

15. Establishment & Certifications Page

This page stores and displays official IIC-related documents.

Content

- Establishment Certificate
- Establishment Order
- IIC Certificates
- Star Rating Certificates
- Awards
- Recognition
- Annual Reports
- NISP Documents

Each document contains:

- Title
 - Category
 - Year
 - Description
 - Thumbnail
 - PDF/document
-

16. Ideas & Queries Page

This is an interactive section of the website.

The page will contain two main options:

 Submit an Idea

 Ask a Query

16.1 Submit an Idea

Users can submit innovative ideas.

Form Fields

- Full Name
- Register Number, optional
- Department
- Year
- Email
- Phone, optional
- Idea Title
- Category
- Problem Statement
- Proposed Solution
- Expected Impact
- Attachment, optional

Idea Status

- Pending
- Under Review
- Approved
- Rejected

The default status is **Pending**.

The administrator can review the idea and update its status.

16.2 Ask a Query

Users can submit questions to the IIC.

Query Categories

- Startup
- Innovation
- Patent/IPR
- Funding
- Event
- General

Form Fields

- Name
- Email
- Department
- Subject
- Category
- Message
- Attachment, optional

Query Status

- Open
- In Progress
- Resolved
- Closed

Administrators can view queries and add replies.

17. Contact Us Page

The Contact page contains:

- IIC email
- Phone number
- Institution address
- Google Map
- Social media links
- Contact form

Contact Form

Fields:

- Name
- Email
- Phone
- Subject
- Message

The submitted message will be stored in Supabase.

18. Admin Authentication

The Admin Dashboard will use **Supabase Authentication**.

Login

Admin enters:

- Email
- Password

After successful authentication:

Login



Supabase Authentication



Verify User



Admin Dashboard

Unauthenticated users cannot access protected admin pages.

19. Admin Dashboard

The Admin Dashboard will provide complete website management.

Admin Dashboard



20. Dashboard Overview

Display statistics such as:

- Total Members
 - Total Events
 - Upcoming Events
 - Total Notices
 - Active Projects
 - Ideas Received
 - Pending Ideas
 - Open Queries
-

21. Admin CRUD Operations

For most modules, administrators can perform:

CREATE
READ
UPDATE
DELETE

Example:

Add Event
View Event
Edit Event
Delete Event

22. Supabase Database Tables

The project will use the following tables:

leadership
members
events
notices
ambassadors
research
projects
certificates
gallery
ideas
queries
contacts
settings

Admin authentication will be managed using **Supabase Auth** rather than storing passwords manually.

23. Database Overview

Leadership

id
name
position
department
image_url
email
linkedin
display_order
is_active
created_at

Members

id
name
position
team
department
image_url
display_order
is_active

created_at

Events

id
title
description
poster_url
event_date
event_time
venue
registration_url
status
created_at

Notices

id
title
description
category
published_date
deadline
attachment_url
external_link
is_pinned
is_active
created_at

Innovation Ambassadors

id
name
department
position
image_url
responsibilities
achievements
year
is_active
created_at

Research

id
title
description
researcher
mentor
category
status
image_url
document_url
created_at

Projects

id
title
description
team_lead
team_members
mentor
technologies
category
status
progress
image_url
project_url
created_at

Certificates

id
title
category
year
description
thumbnail_url
document_url
created_at

Ideas

id
name
register_number
department

year
email
phone
title
category
problem_statement
proposed_solution
expected_impact
attachment_url
status
admin_remarks
created_at

Queries

id
name
email
department
subject
category
message
attachment_url
reply
status
created_at

Contacts

id
name
email
phone
subject
message
created_at

24. Supabase Storage

Storage buckets can include:

leadership-images

member-images
event-posters
event-gallery
ambassador-images
project-images
research-files
certificates
idea-attachments
query-attachments
website-assets

The database stores the file URL while the actual file is stored in Supabase Storage.

25. Security

The project should use:

- Supabase Authentication
- Row Level Security (RLS)
- Protected Admin Routes
- Input validation
- File type validation
- File size restrictions
- Secure environment variables

Public Access

Public users can:

- Read published website content.
- Insert ideas.
- Insert queries.
- Insert contact messages.

Admin Access

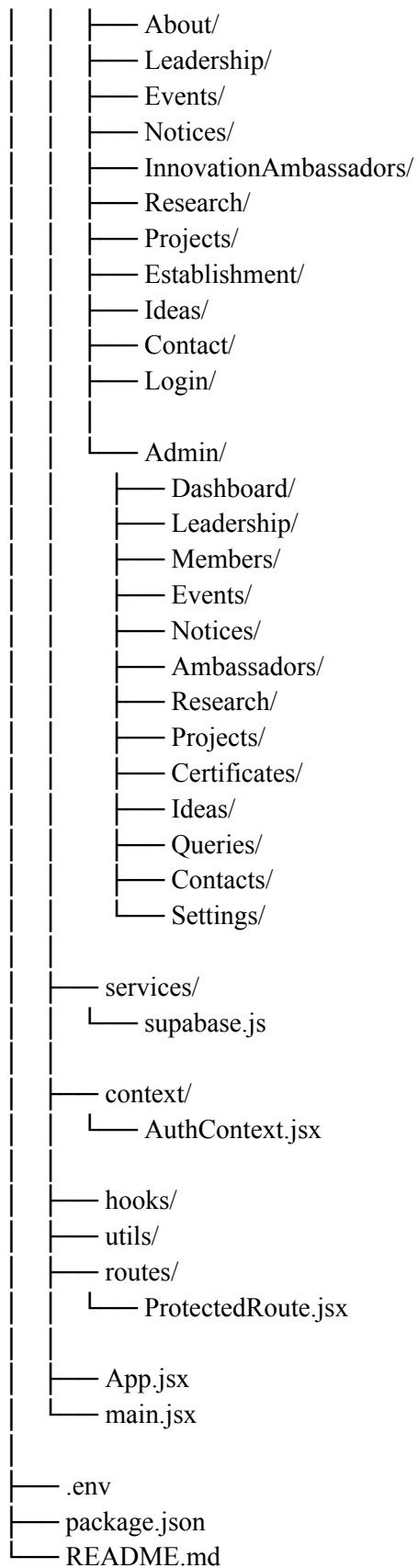
Only authenticated administrators can:

- Add content.
- Edit content.
- Delete content.

- Review ideas.
 - Reply to queries.
 - Access private submissions.
-

26. Project Structure

```
iic-portal/  
├── public/  
├── src/  
│   ├── assets/  
│   ├── components/  
│   │   ├── common/  
│   │   │   ├── Button.jsx  
│   │   │   ├── Loader.jsx  
│   │   │   └── Modal.jsx  
│   │   ├── layout/  
│   │   │   ├── Navbar.jsx  
│   │   │   ├── Footer.jsx  
│   │   │   └── AdminSidebar.jsx  
│   │   ├── cards/  
│   │   │   ├── MemberCard.jsx  
│   │   │   ├── EventCard.jsx  
│   │   │   ├── NoticeCard.jsx  
│   │   │   └── ProjectCard.jsx  
│   │   ├── forms/  
│   │   │   ├── IdeaForm.jsx  
│   │   │   ├── QueryForm.jsx  
│   │   │   └── ContactForm.jsx  
│   │   └── Countdown/  
│   │       └── Countdown.jsx  
│   └── pages/  
│       └── Home/
```



27. Public Routes

/	Home
/about	About
/leadership	Leadership & Members
/events	Events
/notices	Notices
/ambassadors	Innovation Ambassadors
/research	Research & Development
/projects	Active Projects
/establishment	Establishment & Certifications
/ideas	Ideas & Queries
/contact	Contact Us

28. Admin Routes

/admin/login
/admin/dashboard
/admin/leadership
/admin/members
/admin/events
/admin/notices
/admin/ambassadors
/admin/research
/admin/projects
/admin/certificates
/admin/ideas
/admin/queries
/admin/contacts
/admin/settings

All routes except [/admin/login](#) must be protected.

29. UI Design Direction

The website should have a **modern innovation/startup ecosystem appearance**, rather than a traditional college website.

Design Style

- Clean white background
- Blue, purple, and cyan accents
- Modern typography
- Large headings
- Soft shadows
- Rounded cards
- Glassmorphism elements
- Smooth animations
- Responsive design
- Interactive hover effects
- Subtle gradients

Suggested Colors

Primary #2563EB
Secondary #7C3AED
Accent #06B6D4
Success #10B981
Background #F8FAFC
Dark #0F172A

30. Responsive Design

The website must support:

- Desktop
- Laptop
- Tablet
- Mobile

The mobile version must include:

- Responsive navigation menu
- Touch-friendly buttons
- Responsive cards
- Optimized images

- Mobile-friendly forms
-

31. Team Division

Team/Role	Responsibility
Project Lead	Architecture, coordination, integration
UI/UX	Design system and page layouts
Frontend Team	React pages and reusable components
Supabase Team	Database, authentication, storage and RLS
Admin Dashboard Team	CRUD and content management
Testing Team	Functional and responsive testing
Content Team	Text, images, events, certificates and member information

32. Development Phases

Phase 1 — Foundation

- Create React project
- Configure Tailwind CSS
- Configure routing
- Create design system
- Build Navbar and Footer
- Configure Supabase

Phase 2 — Public Website

Develop:

- Home
- About
- Leadership & Members
- Events
- Notices

- Innovation Ambassadors
- R&D
- Active Projects
- Establishment & Certifications
- Ideas & Queries
- Contact

Phase 3 — Supabase Integration

- Create database tables
- Configure storage
- Configure authentication
- Configure RLS policies
- Connect React forms

Phase 4 — Admin Dashboard

Develop CRUD management for all dynamic website content.

Phase 5 — Testing

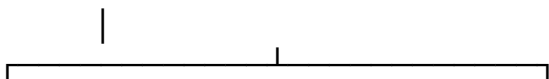
- Functional testing
- Form testing
- Authentication testing
- Responsive testing
- Browser testing
- Security testing

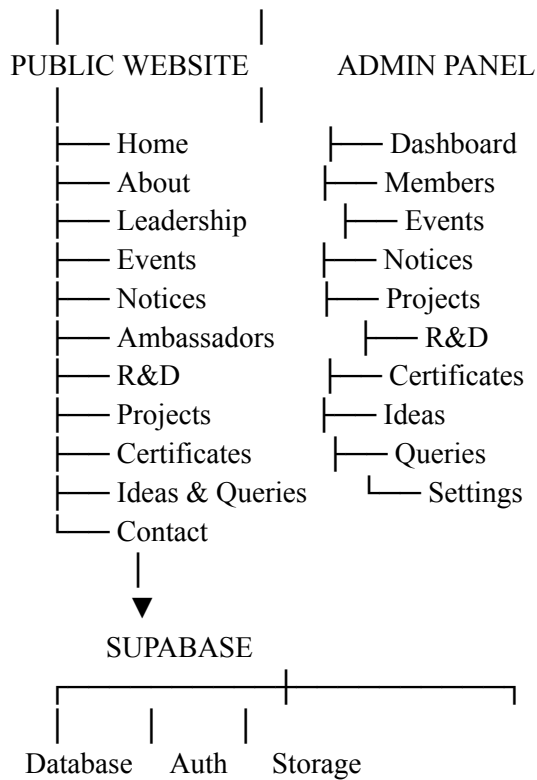
Phase 6 — Deployment

- Deploy React application
- Configure environment variables
- Connect production Supabase project
- Test production website

33. Final Architecture

IIC PORTAL





34. Final Project Summary

The **IIC Portal** will be a dynamic, modern, and fully manageable institutional platform built using:

React + Vite + Tailwind CSS + Supabase

The project does **not require a traditional backend server**. Supabase will handle the database, admin authentication, file storage, and access control.

The final system will provide:

- A modern public website
- Dynamic event management
- Upcoming event countdowns
- Notice publishing
- Leadership and team management
- Innovation Ambassador profiles
- R&D showcase
- Active project showcase
- Certifications and document management
- Public idea submission

- Public query submission
- Contact management
- Secure Admin Dashboard
- Full content management through Supabase